MODULE 1 · LECTURE 14

Smartphone Applications in Agriculture: Market Price Tracking, Farm Advice and Post-harvest Management

How farm apps work, the Indian apps that give market prices, weather-based and expert advice and pest diagnosis, how apps support storage, credit and sale after harvest, and how to judge whether an app is trustworthy.

Learning objectives

After this lecture you should be able to:

1. Describe the smartphone features and types of apps used in agriculture.
2. Explain how a farm app obtains and presents data.
3. Use market price information to compare mandis and calculate net returns.
4. Describe Indian apps for weather-based advice, expert advice and pest diagnosis.
5. Explain how apps and portals support storage, warehouse receipts, credit and sale after harvest.
6. Evaluate a farm app for reliability, usability and data safety.

Scope: Lecture 7 introduced the Internet and e-NAM, Lecture 10 the role of ICT in farming, and Lecture 11 weather forecasts and agromet advisories. This lecture looks at the smartphone apps that bring these services to farmers. Geospatial apps follow in Lecture 15, decision support systems in Lecture 16, and AI-based pest detection in Lecture 25.

1. Why smartphones matter in agriculture

In India, almost all Internet use is now through mobile devices: at the end of June 2026, about 94% of Internet subscriptions were used through mobile devices, and average wireless data use was 28.39 GB per subscriber per month (Telecom Regulatory Authority of India [TRAI], 2026). A smartphone combines a camera, satellite positioning, sensors and a connection in one pocket device, so the same tool can check prices, receive weather warnings, photograph a diseased leaf, record farm operations and receive payments. Studies of mobile-based advisory services show modest gains in yields and adoption of recommended practices at very low cost per farmer (Fabregas et al., 2019).

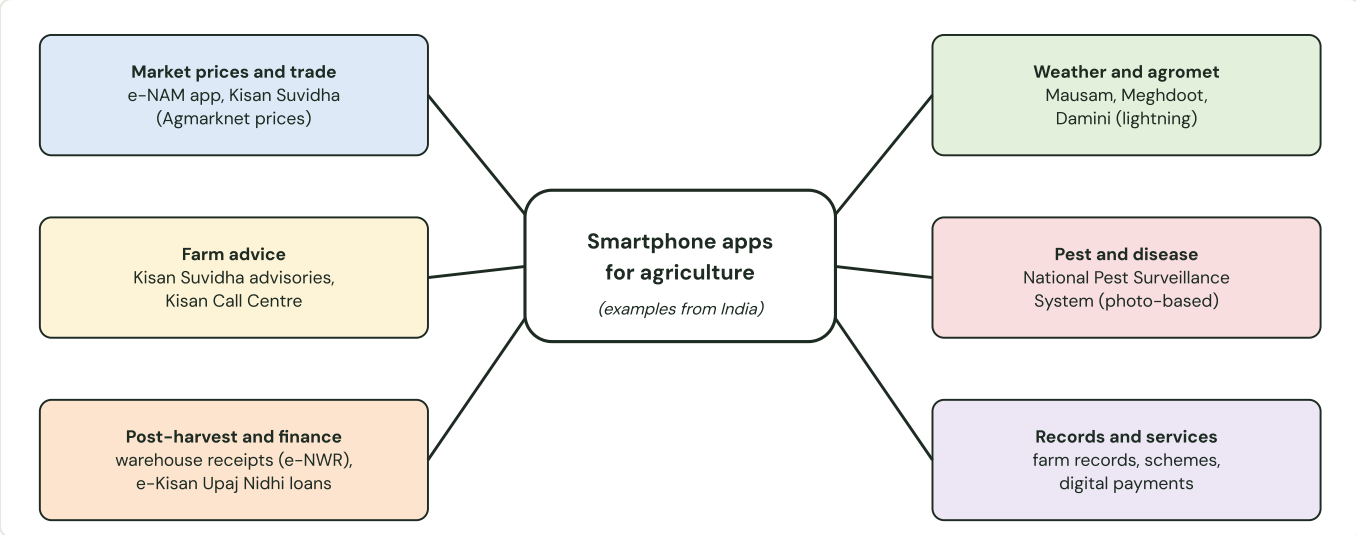


Figure 1. Main types of farm apps, with Indian examples discussed in this lecture. Original diagram.

2. Smartphone features and types of apps

Smartphone feature	Agricultural use
Touch screen, icons and voice	Easy use with little typing; voice input and audio for farmers who read with difficulty
GPS (satellite positioning)	Location-specific weather and advisories; nearest mandi, dealer or warehouse; mapping field boundaries
Camera	Photographing pests, diseases and nutrient deficiencies for diagnosis; scanning QR codes on products
Mobile data and Wi-Fi	Downloading prices, forecasts and videos; uploading photos and records
Bluetooth	Reading data from nearby sensors and devices
Notifications	Alerts for extreme weather, lightning, pest outbreaks and payment receipts
Storage	Offline use of content where the network is weak

Type of app	Description	Notes
Native app	Installed from an app store and written for a particular operating system (e.g. Android, Lecture 4)	Fast; uses all phone features; can work offline
Web app	Runs inside the browser; nothing to install	Works on any phone; needs a connection
Hybrid app	Web technology packaged as an installable app	One code base for several systems
SMS, IVR and messaging services	Text or voice services that also reach basic phones	Important backup where smartphones or data are lacking

How a farm app gets its data

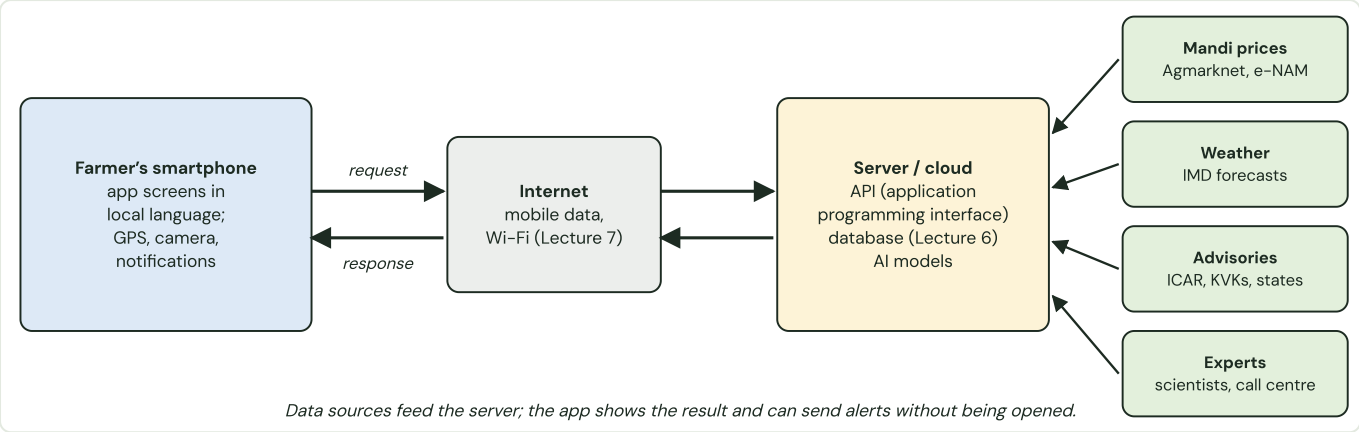


Figure 2. How a farm app works: the phone requests data from a server, which draws on price, weather, advisory and expert sources. Original diagram.

Most farm apps are **clients** of a server (Lecture 7). The app sends a request — for example, today's onion prices near the farmer's location — to an **API** (application programming interface) on a server, which reads a **database** (Lecture 6) and returns the answer. Government data are increasingly published as open data: daily mandi prices collected through AGMARKNET are released as a dataset on the Open Government Data platform, from which developers can build services (Open Government Data Platform India, n.d.).

3. Market price tracking

Price reports from regulated markets usually give the **arrival** (quantity brought to the market) and three prices per quintal: the **minimum**, the **maximum** and the **modal price** — the price at which most of the produce was sold. The modal price is the best guide to what a farmer is likely to receive.

App or service	What it offers
e-NAM mobile app	Launched in October 2016 alongside the e-NAM trading portal; as of December 2025 it offered free price information for 247 notified commodities and access to online trading. Its "Platform of Platforms", introduced on 14 July 2022, links users to logistics, warehousing, quality assurance, grading and packaging, inputs, finance, insurance and advisory services (Press Information Bureau [PIB], 2026)
Kisan Suvidha	An omnibus government app giving market prices in the nearest area and the maximum price in the state and in India, along with weather, dealers, advisories and plant protection (Directorate of Plant Protection, Quarantine & Storage [PPQS], n.d.)
AGMARKNET data	Daily arrivals and prices from regulated markets, used by many government and private apps

Worked example 1 — choosing a mandi using app prices (illustrative values)
A farmer has 40 quintals of wheat. The app shows modal prices: Mandi A ₹2,300/q (8 km), Mandi B ₹2,420/q (35 km), Mandi C ₹2,480/q (70 km). Transport costs ₹1.5 per quintal per km and market charges are 2% of the price.
Net price = modal price - transport - charges:
A: 2,300 - 12.0 - 46.0 = ₹2,242/q → ₹89,680
B: 2,420 - 52.5 - 48.4 = ₹2,319/q → ₹92,764
C: 2,480 - 105.0 - 49.6 = ₹2,325/q → ₹93,016
Mandi C gives only ₹252 more than Mandi B for twice the distance, so the extra time, fuel risk and possible price change on arrival may favour Mandi B. The highest displayed price is not always the best choice.

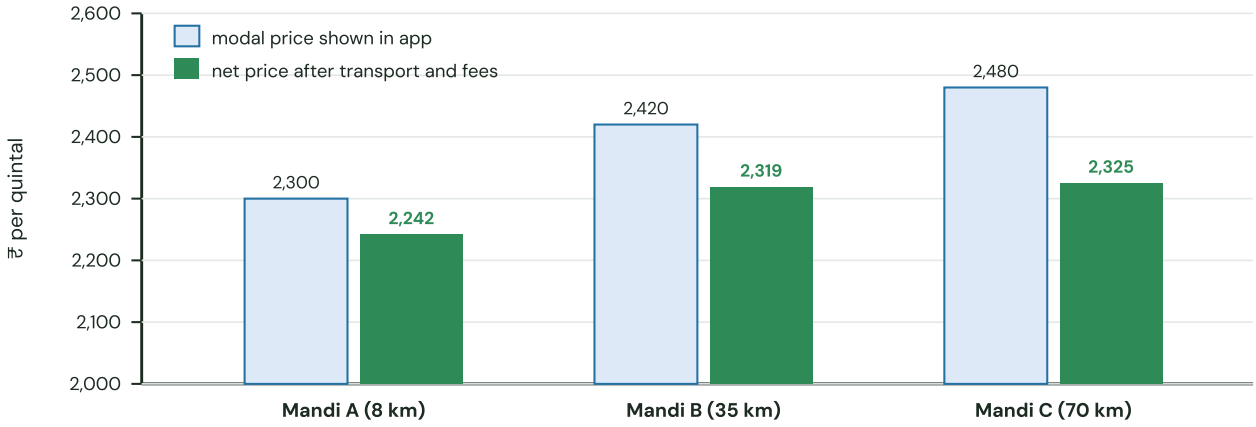


Figure 3. Modal price shown in the app and net price after transport and market charges, from worked example 1. Original chart.

4. Farm advice

Weather and agromet apps

App	Developed by	What it provides
Mausam	Ministry of Earth Sciences / IMD	Observed weather, forecasts and radar images, with warnings of impending weather events (Government of Goa, 2023)
Meghdoot	Joint initiative of IMD, IITM and ICAR; app developed by ICRISAT	District-wise crop and livestock advisories issued every Tuesday and Friday, based on past and forecast weather — for decisions on sowing, spraying, fertiliser, irrigation and animal vaccination (Government of Goa, 2023; International Crops Research Institute for the Semi-Arid Tropics [ICRISAT], 2022)
Damini	IITM Pune (Ministry of Earth Sciences)	Location of current lightning and the direction of thunderstorms, from IITM's lightning location network (Government of Goa, 2023)

Meghdoot delivers the GKMS agromet advisories described in Lecture 11; Damini alerts are a safety tool — farm workers in open fields are at high risk during thunderstorms.

Expert advice and multi-service apps

- **Kisan Suvidha** gives weather for the current day and the next 5 days, extreme-weather alerts, input dealers, market prices, agro-advisories, plant protection and integrated pest management (IPM) practices in one app (PPQS, n.d.).

- **Kisan Call Centre** (1800-180-1551, Lecture 7) remains the voice route to experts, and many apps link to it.
- State agriculture departments, agricultural universities and KVKs publish crop-specific apps and video advice in local languages.

Pest and disease diagnosis

In August 2024 the Ministry of Agriculture and Farmers Welfare launched the **National Pest Surveillance System (NPSS)**, an AI-based platform through which farmers photograph pest-infested crops or insects, upload the photos and receive advice on identification and control from experts. The Minister presented it as a way to reduce farmers' dependence on pesticide retailers (Athrady, 2024). Image-based diagnosis is examined further in Lecture 25.

> **Taking a useful photo for diagnosis:** photograph the affected part in daylight and in focus, include both sides of the leaf, add one picture of the whole plant and one of the field, and note the crop, variety, growth stage and recent sprays.

5. Post-harvest management

Prices are often lowest just after harvest, when everyone sells at once. Storing the crop and selling later can pay — if the farmer can meet immediate cash needs and the produce is stored safely. Digital systems now link storage, quality, credit and sale.

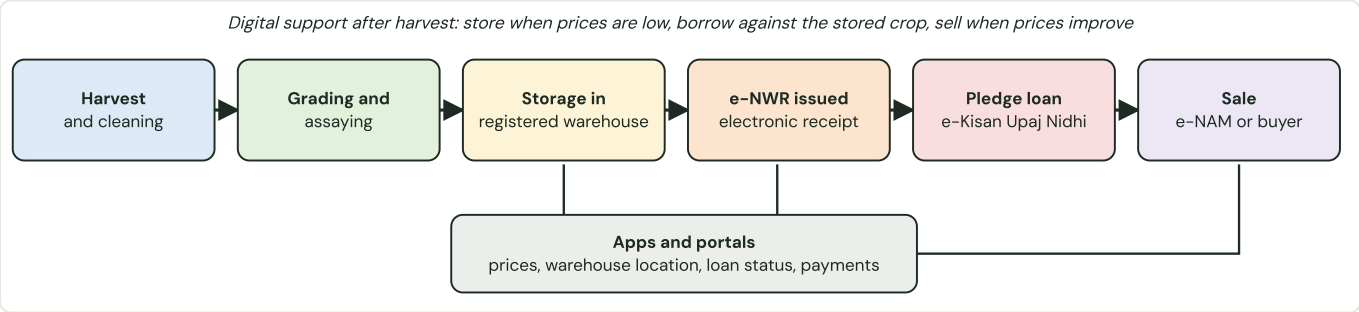


Figure 4. Digital support for storage, credit and sale after harvest. Original diagram.

- **Electronic negotiable warehouse receipts (e-NWRs)** — since the Warehousing (Development and Regulation) Act, 2007, warehouse receipts are negotiable instruments. e-NWRs issued by warehouses registered with the Warehousing Development and Regulatory Authority (WDRA) assure banks that the stored commodity exists and has the stated quality, so they serve as collateral (WDRA, n.d.).
- **e-Kisan Upaj Nidhi** — WDRA's online gateway through which farmers apply for loans against their e-NWRs. The margin is usually 25% of the value, and farmers with a Kisan Credit Card can obtain pledge loans at 7% a year for up to six months, subject to the interest subvention available (WDRA, n.d.).
- **e-NAM's Platform of Platforms** connects users to logistics, warehousing, assaying, grading and packaging services through the e-NAM app (PIB, 2026).
- Other apps help farmers find cold stores and transport, check grading standards, and trace produce with QR codes.

Worked example 2 — store and borrow, or sell now? (illustrative prices and costs; loan terms as described by WDRA)
Wheat 50 q; price now ₹2,400/q (value ₹1,20,000); expected price after 4 months ₹2,700/q.
Loan against the e-NWR at 25% margin = 75% × 1,20,000 = ₹90,000 → interest at 7% for 4 months = 90,000 × 0.07 × 4/12 = ₹2,100.
Storage charges ₹8/q per month → 50 × 8 × 4 = ₹1,600. Storage loss 1% → 49.5 q left.
Sale after 4 months: 49.5 × 2,700 = ₹1,33,650. Gain over selling now = 1,33,650 − 1,20,000 − 1,600 − 2,100 = **₹9,950**.
Break-even price = (1,20,000 + 1,600 + 2,100) ÷ 49.5 = **₹2,499/q** — storing pays only if the later price exceeds this. Meanwhile the loan gives the farmer ₹90,000 in cash at harvest.

6. Choosing and using farm apps wisely

Question to ask	Why it matters
Who publishes the app?	Prefer government, university, ICAR or well-known organisations; check the developer name in the app store
Where do its data come from, and how recent are they?	Prices and forecasts must be current and from a stated source
Is it in my language, and does it work with a weak signal?	Local language, voice, images and offline mode widen use
What permissions does it ask for?	A price app does not need contacts or messages; refuse unnecessary permissions
How does it use my personal data?	Read the privacy notice; personal data are protected under the DPDP Act, 2023 (Lecture 6)
Does it give a way to contact an expert?	Advice should be checkable — call centre, KVK or extension officer
Is it genuine?	Fake apps and links can steal money or data (Lecture 7); install only from the official store

Benefits of farm apps	Limitations
Quick access to prices, weather and advice	Need a smartphone, data and some digital skills
Better marketing decisions and less dependence on intermediaries	Displayed prices may change by the time produce reaches the market
Early warnings of weather and pests	Generic advice may not fit local conditions
Records for credit, insurance and schemes	Too many separate apps; data not always shared between them
Easier access to storage, credit and payments	Privacy and fraud risks

7. Key takeaways

- Smartphones are now the main route to the Internet in India; their GPS, camera, sensors and notifications make location-specific farm services possible.
- Farm apps are clients of servers that combine price, weather, advisory and expert data through APIs and databases.

- Market apps report arrivals and minimum, maximum and modal prices; the best market is the one with the highest net price after transport and charges.
- The e-NAM app (2016) gives prices for 247 commodities and access to trading and the Platform of Platforms; Kisan Suvidha combines prices, weather, dealers and advice.
- Mausam, Meghdoot and Damini deliver forecasts, agromet advisories and lightning alerts; NPSS (2024) offers photo-based pest identification and advice.
- e-NWRs from WDRA-registered warehouses, with loans through e-Kisan Upaj Nidhi, let farmers store, borrow and sell later; storage pays only above the break-even price.
- Choose apps from trusted publishers, check their data sources and permissions, and protect personal data.

8. Examination questions

Short-answer questions (2 marks)

Q1. Name four smartphone features useful in agriculture.

GPS, camera, mobile data connection and notifications (also touch screen, voice and Bluetooth).

Q2. Distinguish a native app from a web app.

A native app is installed and written for one operating system; a web app runs in the browser without installation.

Q3. What is the modal price?

The price at which most of the produce was sold in a market on a day.

Q4. When was the e-NAM mobile app launched, and how many commodities does it give prices for?

October 2016; 247 notified commodities (December 2025).

Q5. What is e-NAM's Platform of Platforms?

A framework (from July 2022) linking e-NAM users to logistics, warehousing, quality, finance, insurance and advisory services.

Q6. List the services in the Kisan Suvidha app.

Weather for today and the next 5 days with alerts, dealers, market prices, agro-advisories, plant protection and IPM practices.

Q7. What is Meghdoot?

An app of IMD, IITM and ICAR (developed by ICRISAT) that delivers district crop and livestock advisories every Tuesday and Friday.

Q8. What does the Damini app do?

Shows the location of current lightning and the direction of thunderstorms.

Q9. What is an e-NWR?

An electronic negotiable warehouse receipt issued by a WDRA-registered warehouse, usable as collateral for loans.

Q10. What is the National Pest Surveillance System?

An AI-based platform launched in 2024 where farmers upload pest photos and receive identification and control advice.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain how a smartphone farm app works, with a diagram.

Smartphone features; client–server, API, database, data sources (Figure 2); native, web and hybrid apps; SMS/IVR backup.

Q2. Describe smartphone applications for market price tracking and show how a farmer should choose a market.

Price terms; e-NAM app and PoP, Kisan Suvidha, AGMARKNET data; worked example 1 and Figure 3.

Q3. Describe smartphone apps that provide weather-based and expert farm advice in India.

Mausam, Meghdoot, Damini (table); Kisan Suvidha; Kisan Call Centre; NPSS photo-based diagnosis.

Q4. Explain the role of apps and portals in post-harvest management.

Figure 4; e-NWR and WDRA; e-Kisan Upaj Nidhi loan terms; e-NAM Platform of Platforms; worked example 2 with break-even price.

Q5. How should a farmer evaluate a farm app? Discuss the benefits and limitations of farm apps.

Evaluation checklist; permissions and DPDP Act; fake apps; benefits and limitations table.

Q6. Discuss the scope of smartphone applications in Indian agriculture.

Figure 1 categories with examples; TRAI mobile-use figures; evidence from Fabregas et al. (2019); limitations.

9. Glossary

Term	Meaning
API	Application programming interface — rules by which one program requests data from another
Arrival	Quantity of a commodity brought to a market on a day
e-Kisan Upaj Nidhi	WDRA's online gateway for loans against e-NWRs
e-NWR	Electronic negotiable warehouse receipt
Hybrid app	Web technology packaged as an installable app
IVR	Interactive voice response — automated phone menus
Margin	Part of the value of pledged goods that the bank does not lend
Modal price	The most common selling price in a market on a day
Native app	An app installed on and written for a particular operating system
Net price	Price received after transport and market charges
NPSS	National Pest Surveillance System
Open data	Data published for anyone to use, often through APIs
Pledge loan	Loan secured by goods held in storage
Platform of Platforms	e-NAM framework linking users to value-chain services
WDRA	Warehousing Development and Regulatory Authority
Web app	An application used through a web browser

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All diagrams and the chart are original. App descriptions are summarised from the publishers' pages and the cited sources; features and coverage change frequently, so check the current version. Prices and costs in the worked examples are illustrative, and every result shown was recalculated.

Practice MCQs

Q1. Which smartphone feature lets an app give advice for the farmer's exact location?

(a) Bluetooth (b) GPS
(c) Camera flash (d) Speaker

Q2. An app that runs in the browser without installation is a

(a) native app (b) web app
(c) hybrid app (d) IVR service

Q3. In mandi price reports, the price at which most of the produce was sold is the

(a) minimum price (b) maximum price
(c) modal price (d) support price

Q4. The e-NAM mobile app was launched in

(a) April 2016 (b) October 2016
(c) July 2022 (d) August 2024

Q5. e-NAM's Platform of Platforms links users to

(a) only weather forecasts (b) logistics, warehousing, quality, finance and other value-chain services
(c) only seed dealers (d) land records

Q6. The Kisan Suvidha app gives weather information for

(a) the current day only (b) the current day and next 5 days
(c) the next month (d) the whole season

Q7. Meghdoot is a joint initiative of

(a) IMD, IITM and ICAR
(b) ISRO and NABARD
(c) WDRA and FCI
(d) TRAI and BSNL

Q8. The Damini app provides

(a) mandi prices
(b) lightning alerts
(c) soil-test results
(d) loan status

Q9. Through the National Pest Surveillance System, farmers

(a) buy pesticides online
(b) upload photos of pest attacks for identification and advice
(c) sell produce
(d) book tractors

Q10. Electronic negotiable warehouse receipts are issued by

(a) any village shop
(b) WDRA-registered warehouses
(c) mandi commission agents
(d) mobile operators

Q11. Loans against e-NWRs usually require a margin of

(a) 5%
(b) 10%
(c) 25%
(d) 50%

Q12. Storing produce is profitable only if the later price exceeds the

(a) modal price today
(b) break-even price including storage, interest and losses
(c) minimum support price
(d) maximum price today

Answers

Q1 — (b) GPS
Q2 — (b) web app
Q3 — (c) modal price
Q4 — (b) October 2016
Q5 — (b) logistics, warehousing, quality, finance and other value-chain services
Q6 — (b) the current day and next 5 days
Q7 — (a) IMD, IITM and ICAR
Q8 — (b) lightning alerts
Q9 — (b) upload photos of pest attacks for identification and advice
Q10 — (b) WDRA-registered warehouses
Q11 — (c) 25%
Q12 — (b) break-even price including storage, interest and losses